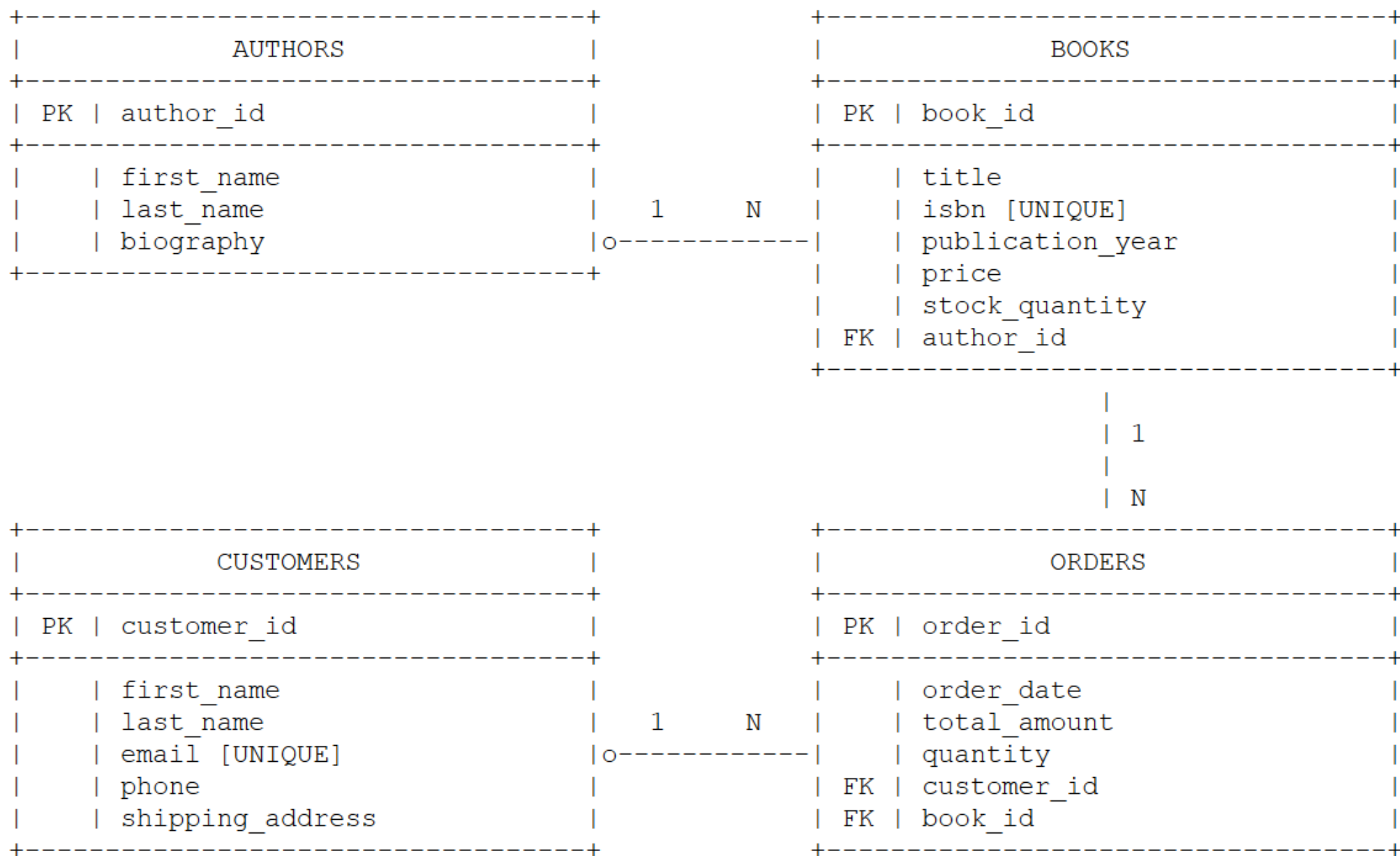


WEEK 5 - EXERCISE

MONDAY

ERD



AUTHORS TABLE (Independen Entity)

- **Attributes:**
 - author_id **[PK]** (Integer, AUTOINCREMENT) → *Unique identifier for each author.*
 - first_name (Text, NOT NULL, CHECK) → *Cannot be null or empty spaces.*
 - last_name (Text, NOT NULL, CHECK) → *Cannot be null or empty spaces.*
 - biography (Text, Nullable).
- **Data Integrity Constraints:**
 - CHECK(length(trim(first_name)) > 0)
 - CHECK(length(trim(last_name)) > 0)
- **Relationship:** Connects to **books** with a **One-to-Many (1:N)** line.

BOOKS TABLE (Dependent on authors)

- **Attributes:**

book_id **[PK]** (Integer, AUTOINCREMENT) → *Unique identifier for each book.*

title (Text, NOT NULL, CHECK) → *Cannot be empty.*

isbn (Text, UNIQUE, NOT NULL, CHECK) → *Must be distinct; prevents duplicate listings of the same edition.*

publication_year (Integer, CHECK) → *Must fit realistic chronological boundaries.*

price (Real, NOT NULL, CHECK) → *Financial value must be positive.*

stock_quantity (Integer, NOT NULL, DEFAULT 0, CHECK) → *Defaults to zero; cannot be negative.*

author_id **[FK]** (Integer, NOT NULL) → *References authors.author_id.*

- **Data Integrity Constraints:**

CHECK(length(trim(title)) > 0)

CHECK(length(trim(isbn)) >= 10)

CHECK(publication_year >= 1455 AND publication_year <= strftime('%Y', 'now'))

CHECK(price > 0.0)

CHECK(stock_quantity >= 0)

- **Referential Integrity Rule:**

ON DELETE RESTRICT → *Prevents deleting an author if they still have books linked to their record.*

- **Relationship:** Connects to **orders** with a **One-to-Many (1:N)** line.

CUSTOMERS TABLE (Independen Entity)

- **Attributes:**

- customer_id **[PK]** (Integer, AUTOINCREMENT) → *Unique identifier for each customer.*
- first_name (Text, NOT NULL, CHECK) → *Cannot be empty.*
- last_name (Text, NOT NULL, CHECK) → *Cannot be empty.*
- email (Text, UNIQUE, NOT NULL, CHECK) → *Must be distinct for logins and follow standard email patterns.*
- phone (Text, Nullable, CHECK) → *Optional, but must meet basic length requirements if provided.*
- shipping_address (Text, NOT NULL, CHECK) → *Required for shipping calculations.*

- **Data Integrity Constraints:**

- CHECK(length(trim(first_name)) > 0)
- CHECK(length(trim(last_name)) > 0)
- CHECK(email LIKE '%_@__%._%') → *Validates basic format string structure (e.g., user@domain.com).*
- CHECK(phone IS NULL OR length(phone) >= 7)
- CHECK(length(trim(shipping_address)) > 0)

- **Relationship:** Connects to **orders** with a **One-to-Many (1:N)** line.

ORDERS TABLE (Dependent transaction entity)

- **Attributes:**

- order_id [**PK**] (Integer, AUTOINCREMENT) → *Unique identifier for each transaction.*
- order_date (Text/Timestamp, NOT NULL, DEFAULT CURRENT_TIMESTAMP) → *Automatically timestamps the record upon generation.*
- total_amount (Real, NOT NULL, CHECK) → *Total financial balance must be non-negative.*
- quantity (Integer, NOT NULL, DEFAULT 1, CHECK) → *Must purchase at least 1 book per transaction.*
- customer_id [**FK**] (Integer, NOT NULL) → *References customers.customer_id.*
- book_id [**FK**] (Integer, NOT NULL) → *References books.book_id.*

- **Data Integrity Constraints:**

- CHECK(total_amount >= 0.0)
- CHECK(quantity > 0)

- **Referential Integrity Rules:**

- FOREIGN KEY (customer_id) REFERENCES customers(customer_id) ON DELETE RESTRICT
- FOREIGN KEY (book_id) REFERENCES books(book_id) ON DELETE RESTRICT

Note: *Both rules guarantee that transactional sales history cannot be broken or orphaned by deleting an active customer or a cataloged book.*

EMPLOYEES TABLE

Column	Data Type	Constraints	Purpose
employee_ID	INTEGER	PRIMARY KEY AUTOINCREMENT	Unique identifier for each staff member.
first_name	TEXT	NOT NULL, CHECK(length(trim(first_name)) > 0)	Prevents empty/blank names.
last_name	TEXT	NOT NULL, CHECK(length(trim(last_name)) > 0)	Prevents empty/blank names.
department	TEXT	NOT NULL, CHECK(department IN ('Marketing', 'Sales', 'Customer Support', 'Engineering'))	Restricts values to valid company departments only.
hired_date	TEXT	NOT NULL	Uses YYYY-MM-DD format (SQLite default).

PRODUCTS TABLE

Column	Data Type	Constraints	Purpose
product_ID	INTEGER	PRIMARY KEY AUTOINCREMENT	Unique identifier for each product.
product_name	TEXT	NOT NULL, CHECK(length(trim(product_name)) > 0)	Descriptive name.
category	TEXT	NOT NULL	Categorizes the item (e.g., Electronics).
price	REAL	NOT NULL, CHECK(price > 0)	Ensures the price is always a positive number.
units_in_stock	INTEGER	NOT NULL, DEFAULT 0, CHECK(units_in_stock >= 0)	Starts at 0; prevents negative inventory.
units_sold	INTEGER	NOT NULL, DEFAULT 0, CHECK(units_sold >= 0)	Tracks total sales count.
discontinued_date	TEXT	NULLABLE	Optional; only populated if the item is removed from catalog.

RECOMMENDED SCHEMA (Spotify)

Column	Data Type	Constraints	Purpose
product_ID	INTEGER	PRIMARY KEY AUTOINCREMENT	Unique identifier for each product.
product_name	TEXT	NOT NULL, CHECK(length(trim(product_name)) > 0)	Descriptive name.
category	TEXT	NOT NULL	Categorizes the item (e.g., Electronics).
price	REAL	NOT NULL, CHECK(price > 0)	Ensures the price is always a positive number.
units_in_stock	INTEGER	NOT NULL, DEFAULT 0, CHECK(units_in_stock >= 0)	Starts at 0; prevents negative inventory.
units_sold	INTEGER	NOT NULL, DEFAULT 0, CHECK(units_sold >= 0)	Tracks total sales count.
discontinued_date	TEXT	NULLABLE	Optional; only populated if the item is removed from catalog.